Top-down shooter (Unity / C#)

Author: Jakub Baumgartner · **Engine:** Unity · **Language:** C# · **Repository:** github.com/j4kub0k/C--zapocet

Functionality

A top-down shooter. The player moves with WASD, aims with the mouse and shoots to defeat all enemies (melee and ranged) and the boss. When no enemies remain, the win UI is shown.

- **Health and ammo limits**, displayed next to the player by the HUD scripts.
- **Weapons:** pistol (limited ammo, slow fire), machine gun (fast fire), shotgun (3-bullet spread). Running out of ammo falls back to the pistol.
- **Enemies:** *Patrol* (waypoints, loop or back-and-forth), *Attack* (melee chase when the player is in range and visible), *RangeAttack* (shoots at intervals within a 60° field of view), *Die* (health and death effect).
- **Power-ups** (one small script each): heal, max-health up, machine gun and shotgun pickups, damage boost, infinite ammo, invincibility, wider vision, faster movement.
- **Save system:** *PLayerData* (serializable snapshot of health, ammo, gun and position) is written to and read from *save.json* in Application.persistentDataPath by the static *save* class using JsonUtility. Saved on quit or on request, loaded on start, deleted on game over or a new game.
- **Menus:** main menu (new game, continue, quit), pause menu (resume, save, load, reset, quit) and a game over screen.

Design notes

Each behaviour is its own MonoBehaviour component (composition over inheritance). The weapons share a *fire(bool infinite)* method, which *controller* calls depending on the selected gun type. Persistence is isolated in the static *save* class and the plain data class *PLayerData*. Code is documented with XML (///) comments on classes and methods.

Repository structure

Path	Content
Assets/scripts/	All game logic (C#): player, weapons, enemies, power-ups, UI, save system
Assets/*.unity	Scenes: Menu, Game, GameOver
Assets/*.prefab, *.mat	Bullets, power-ups, doors, materials
Assets/Plugins/Demigiant/	Third-party DOTween library (camera shake), not my code
Assets/*.mp3, *.png, M1911.fbx	Audio, images and weapon model
Packages/, ProjectSettings/	Unity package manifest and project settings
.gitignore	Unity ignore rules (Library/, obj/, Logs/, .vs/ are not versioned)

Build

Open the project in Unity, then File → Build Settings → Build (scenes: Menu, Game, GameOver).